Applied astrodynamics

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1 Introduction

The discipline dealing with engineering applications of celestial mechanics was defined as astrodynamics by Herrick in his renowned book [1]: "It's an engineer specific skill, since the orbit designer has to evaluate various solutions both for the implementation of an orbit, and for its variations, as well as to determine it and find out the best out of the possible solutions, through a process of optimization." Many authors [2–8] agree on emphasizing the typical engineering aspect of astrodynamics, as well as the mathematical: "and mathematical-physical aspects of celestial mechanics closer to the astronomy's domain." Kaplan defines astrodynamics as "the study of controlled trajectories of artificial satellites," thus highlighting the close interrelation that such a discipline has with control and implementation.

Astrodynamics is based on the principles of celestial mechanics. It deals with the study of the dynamics of man-made vehicles, playing with maneuvers and control techniques to achieve specific mission goals. In this way, astrodynamics fulfills the important task of integrating many heterogeneous—yet closely related—elements, such as, for instance, orbit, stability, control, propulsion, and systems. In the past, the definition of space flight dynamics was used, although orbiting bodies move in *free fall* rather than defeating gravitational pull/attraction due to aerodynamic forces. There is still a certain amount of resistance to the name astrodynamics. It is preferable to leave the ascent stage (more properly the subject of launcher flight mechanics) and the reentry phase to the discipline of space flight dynamics, and to assign to astrodynamics the study of dynamics and orbital control and stability/attitude of those vehicles whose sustenance/airworthiness does not necessarily depend on the atmosphere.

The main objectives of astrodynamics are

- Mission analysis, with the study of optimum flight paths and of maneuvers and control techniques, both active and passive, orbital and positioned, to produce them (acquisition and maintenance).
- The study of guidance, navigation, and control systems.
- Orbit and position determination for spacecraft/aerospace vehicles (satellites and probes).
- Orbital propagation for determining the ephemeris of spacecraft/aerospace vehicles.

Moreover, the methodology used to study astrodynamics problems has two different approaches: one is more mathematical and the other is an *engineering approach* that

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allows to *see* the practical results besides being able to formulate them. Applied astrodynamics is therefore a transition from theory to practice in the field of astrodynamics. This chapter aims to present both approaches, while keeping an eye on the peculiarities of CubeSat missions. In fact, if the classical formulation of the two-body problem does not change from one kind of satellite to another, other aspects like perturbations can be characterized to the case of CubeSats. Moreover, advanced astrodynamics addressing low-thrust trajectories are gaining more relevance nowadays for the implementation of interplanetary CubeSat missions.

This chapter is organized as follows: the basics of classical astrodynamics, including the two-body problem, and the definition of orbits are presented in Section 2; the effect of the most relevant environmental perturbations for CubeSat missions are summarized in Section 3; and a selection of advanced topics in astrodynamics, such as gravity assist (GA) maneuvers and invariant manifolds, is presented in Section 4.

2 Principles and laws of astrodynamics

The motion of the planets was deduced by Kepler through several observations carried out by the Danish astronomer Tycho Brahe. The empirical laws describing the motion of the planets, called Kepler laws, can be summarized as:

1. The orbits of the planets are ellipses and the Sun is at one focus.
2. The areas swept by the vector going from the Sun to a planet are proportional to the time necessary to cover/ride them.
3. The squares of the periods necessary for the planets to go through their orbits are proportional to the cube of the major semiaxis.

Kepler's laws were formulated at the beginning of the 17th century based on astronomical observations. A few decades later, Newton developed the mathematical model that supported Kepler's deductions. The basis of this model are the laws of classical mechanics, which can be summarized in the relation:

$$\vec{F} = m\vec{a} \tag{1}$$

where acceleration is considered as an effect of the force causing motion, contrary to Aristotelic physics. The classical formulation of the two-body problem made by Newton assumes a very simple model for the gravitational field of the Earth, yet it reasonably approximates the main features of the orbital motion of a spacecraft in many circumstances. The trajectories obtained with this model follow Kepler's laws and are periodic and return to the same point every period: they are called orbits.

Taking into account the effects, called perturbations, of the other relevant forces in the orbital model gives a more complete representation of the motion of a spacecraft and, at the same time, allows to better design the mission. In this section, it will be shown how to derive a basic description of orbits from the principles of physics and mechanics.

2.1 The two-body problem

The fundamental law of dynamics expressed in Eq. (1), also called law of inertia, is identical in all references and moves by uniform rectilinear motion with respect to the *fixed stars*. Infinite references (Cartesian sets of three) then exist, equivalent to whom the law of inertia is applicable. The force $\vec{F}$, which characterizes astrodynamics, is the force due to Newton's law of universal gravitational attraction exchanged by two bodies with masses m and M placed at a distance r:

$$\vec{F} = \frac{GmM}{r^2} \tag{2}$$

where G is the universal gravitational constant. The constant of proportionality between force and acceleration in Eq. (1) is called inertial mass, while the mass in the expression of the gravitational law Eq. (2) is called gravitational mass. The principle of equivalence states that the ratio between inertial mass and gravitational mass is identical for all bodies, so the inertial mass can be considered equal to the gravitational mass. It follows that the motion of material bodies in free fall is independent of their composition and structure as was highlighted by Galilei through the famous experiments "thought of, but not carried out" throwing different objects (lead and feather) out of Pisa's tower and noticing that they touched the ground at the same moment. Therefore the motion of material bodies subjected only to the gravitational force (Eq. 2) does not depend on their mass.

The so-called two-body or Kepler's problem is the study of the motion of a spacecraft with mass m around a celestial body with mass M ($m \ll M$), assuming a spherically symmetric gravitational field centered in the celestial body and no other forces acting on the system. The acceleration $\ddot{\vec{r}}$ of the spacecraft due to the gravity force $\vec{F}_g$ is found by Newton's law of gravity in what is called the equation of motion of a spacecraft:

$$\frac{d^2\vec{r}}{dt} = \frac{\vec{F}_g}{m} = -\frac{GM}{r^3}\vec{r} = -\frac{\mu}{r^3}\vec{r} \tag{3}$$

where μ is the gravitational parameter (depending on the mass of the celestial body) and $\vec{r}$ is the vector from M to m. Another important quantity to be defined is the angular momentum $\vec{h} \equiv \vec{r} \times \frac{d\vec{r}}{dt}$ of the orbit, which is constant since

$$\vec{r} \times \frac{d^2\vec{r}}{dt} = \vec{r} \times \frac{d^2\vec{r}}{dt} + \frac{d\vec{r}}{dt} \times \frac{d\vec{r}}{dt} = \frac{d}{dt}\left(\vec{r} \times \frac{d\vec{r}}{dt}\right) = \frac{d\vec{h}}{dt} = 0 \tag{4}$$

This result shows that, at every time instant, the position and velocity vectors lie on the same plane, the *orbital plane*, with the angular momentum being perpendicular to it. Therefore the trajectory of the satellite is planar and lies always in the orbital plane.

The equation of motion can be further managed to obtain an insight into the shape of the orbit. Taking the cross-product with the angular momentum yields

$$\vec{h} \times \frac{d^2 \vec{r}}{dt} = -\frac{\mu}{r^3} \vec{h} \times \vec{r} = -\frac{\mu}{r^3} \left(\vec{r} \times \frac{d \vec{r}}{dt} \right) \times \vec{r} = -\mu \frac{d}{dt} \left(\frac{\vec{r}}{r} \right) \tag{5}$$

Integrating the left- and right-hand site of the equation yields

$$\vec{h} \times \frac{d \vec{r}}{dt} = -\mu \frac{\vec{r}}{r} - A \tag{6}$$

where A is a vector constant of the integration depending on the initial position and velocity. Taking the scalar product by r results in

$$\left(\vec{h} \times \frac{d \vec{r}}{dt} \right) \cdot r = -\mu \frac{\vec{r}}{r} \cdot r - A \cdot r \tag{7}$$

Using the property $(a \times b) \cdot c = -(c \times b) \cdot a$ and defining the *true anomaly* ν as the angle on the orbital plane between $\vec{r}$ and A, an equation for the position of the satellite is obtained:

$$h^2 = \mu r + A r \cos \nu \tag{8}$$

$$r = \frac{p}{1 + e \cos \nu} \tag{9}$$

where $p \equiv h^2/\mu$ is a geometrical constant called the semilatus rectum of the orbit and $e \equiv A/\mu$ is another constant called the eccentricity. This expression of the position of the satellite corresponds to the general equation of a conic section in polar coordinates, suggesting that the orbital trajectory in the two-body problem is always a conic section.

Depending on the value of e, the orbit will be a closed or an open trajectory. The type of conic section according to the value of e is described in Table 1.

Table 1 Conic sections of an orbit.

Eccentricity	**Conic section**	**Open/closed trajectory**
$e = 0$	Circle	Closed
$0 < e < 1$	Ellipse	Closed
$e = 1$	Parabola	Open
$e > 1$	Hyperbola	Open

2.2 Energy and orbital period

Eq. (4) can be used to find a relationship between the velocity and the radius of the orbit. Dot multiplying Eq. (1) by $\frac{d\vec{r}}{dt}$ yields

$$
\begin{aligned}
&\frac{d\vec{r}}{dt}\cdot\frac{d^2\vec{r}}{dt}=-\frac{\mu}{r^3}\frac{d\vec{r}}{dt}\cdot\vec{r}\\
&\frac{1}{2}\frac{d}{dt}\left(\dot{\vec{r}}\cdot\dot{\vec{r}}\right)=-\frac{\mu}{r^3}\left(\dot{\vec{r}}+\vec{\omega}\times\vec{r}\right)\cdot\vec{r}\\
&\frac{1}{2}\frac{d}{dt}\left(v^2\right)=-\frac{\mu}{r^2}\dot{r}\\
&\frac{1}{2}\frac{d}{dt}\left(v^2\right)=\frac{d}{dt}\left(\frac{\mu}{r}\right)\\
&\frac{1}{2}v^2-\frac{\mu}{r}=\text{const}
\end{aligned}
\tag{10}
$$

The constant obtained by the integration in the last equation is the specific mechanical energy E associated with the orbital motion. The two terms, in fact, correspond to the kinetic and potential energy of the satellite, respectively:

$$E=\frac{1}{2}v^2-\frac{\mu}{r} \tag{11}$$

2.3 Keplerian elements

As explained in the previous section, in the two-body problem formulation the orbit of a spacecraft around a celestial body is a planar trajectory completely determined by the position and velocity vectors. Another way of representing orbits is through the use of six quantities, called *Keplerian* or *orbital elements*. Since the dimension of this vector of orbital elements is equal to the dimension of the position and velocity vectors, the two representations are equivalent. The advantage of using the orbital elements is that they allow a prompter visualization of the orbit of a satellite rather than the position and velocity vectors.

There exist different sets of elements, the most classical being composed by

1. *Semimajor axis a* of the orbit.
2. *Eccentricity e* of the orbit.
3. *Inclination i* of the orbital plane with respect to a reference plane, e.g., the equatorial plane.
4. *Longitude of the ascending node Ω*, the angle in the reference plane between the intersection with the orbital plane (*line of nodes*) and the first axes of the coordinate system centered in the celestial body.
5. *Argument of periapsis ω*, the angle in the orbital plane between the line of nodes and the periapsis point.
6. *Time of periapsis passage t*.

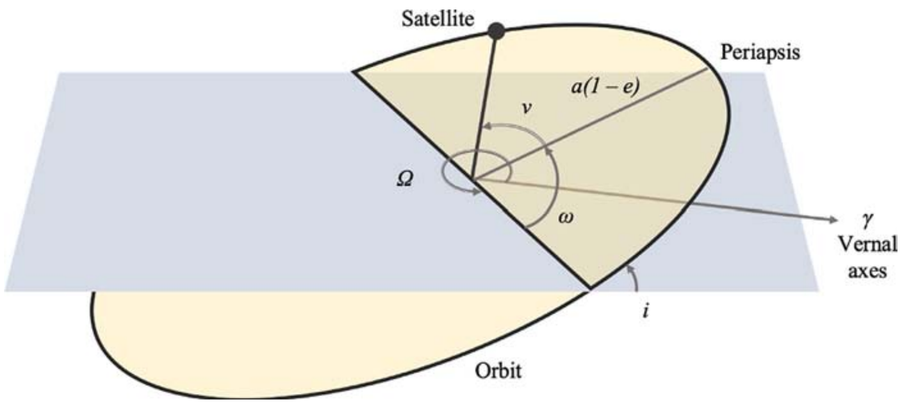

Fig. 1 Keplerian elements.

The first two parameters define the size and shape of the orbit; i, Ω, and ω define its orientation in space; and the sixth parameter defines the position of the spacecraft along the orbit. While it is always possible to calculate position and velocity vectors from the Keplerian elements, the reverse might not be feasible in some cases. For example, for circular orbits, ω is not defined and the calculation of e might cause numerical problems (Fig. 1).

2.4 Orbit classification

The definition of the orbital elements given in the previous section allows a taxonomy for orbits to be defined. In fact, orbits can be classified in accordance with their altitude (related to the semimajor axis a), eccentricity, and inclination. The Earth orbits whose maximum distance from the center of mass of the Earth is smaller than 2000 km are called low Earth orbits (LEO); medium Earth orbits (MEO) are between 2000 and 20,000 km; and high Earth orbits (HEO) are those above 20,000 km. Classification according to eccentricity distinguishes among closed and open orbits: circular ($e=0$) and elliptical ($0<e<1$) orbits are closed trajectories, while parabolic ($e=1$) and hyperbolic ($e>1$) orbits are open trajectories that can escape the gravitational pull of the celestial body. Finally, according to the inclination, orbits can be defined as equatorial ($i=0$ degrees) or polar ($i=90$ degrees).

Combining some of these characteristics yields some of the most important orbits. The orbits with a period equal to a sidereal day are called geosynchronous. Geostationary orbits are geosynchronous circular orbits with zero inclination. A geostationary satellite will appear fixed in the sky to an observer on the Earth and this is why these orbits are used for important telecommunication missions.

CubeSats are not likely to be launched in geostationary orbits, but most of them have been launched in another kind of synchronous orbit, called the Sun-synchronous orbit. In a Sun-synchronous orbit the orbital plane rotates to maintain a constant orientation with respect to the Sun through the year ($\dot{\Omega}=360$ degrees/year), as shown in Fig. 2. At every time of the year (points 1, 2, and 3), the spacecraft will pass over the

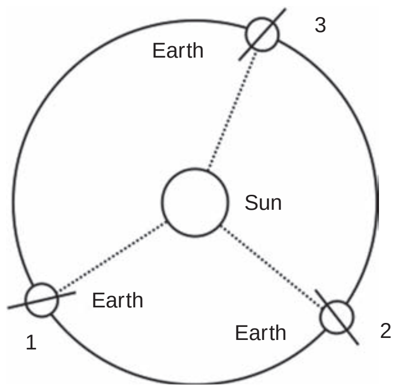

Fig. 2 Orientation of the orbital plane in a Sun-synchronous orbit.

Earth always at the same local solar time. This is very important for remote sensing missions, because at each passage the Earth surface is in similar lightning conditions. The rotation of the orbital plane (*precession*) can be obtained at no cost by exploiting the J_2 perturbation (see next section). To do so, a precise combination of altitude and inclination is required. For low orbits, the most accessible for CubeSats, the resulting inclination is near 90 degrees. As a result of this, a satellite in a low Sun-synchronous orbit spans across the entire surface of the Earth many times per day, with many opportunities for contact with ground stations. This is fundamental, especially if there is a single station that can operate the spacecraft, as in the case of many university missions.

3 Perturbations

A perturbation is a variation of the force model assumed in the unperturbed Keplerian motion. Apart from the gravitational force, in fact, other actions also contribute to determine the motion of a satellite, such as aerodynamic forces and interactions with space radiations. Even the assumptions made on the gravitational field are not adequate to describe the real situation: first of all, the spacecraft and the celestial body are not an isolated system. There are other bodies (e.g., Earth, Sun, Moon, and so on) that contribute to create the gravitational field in which the spacecraft's motion takes place; furthermore, the actual gravitational field of each of these bodies is quite complex, rather than spherically homogeneous.

The perturbed orbits could depend on the mass of the body, therefore a difference between a CubeSat and a larger satellite could exist. This section will deal with the perturbations that are most relevant for CubeSat missions, such as the anomalies of the Earth gravitational field and the atmospheric drag. These actions are typically more relevant at low altitudes, where most of the CubeSat missions take place.

3.1 Anomalies of the Earth gravitational field

The Earth is not a perfect sphere and its mass is not homogeneously distributed inside it. Some portions of the Earth are more *massive*, and some are less. This is the reason why the Earth gravitational field is not uniform in intensity and direction. A realistic model for the Earth gravitational field is derived through the gradient of the gravity potential $\vec{U}$:

$$\ddot{\vec{r}} = \nabla \vec{U}, \quad \vec{U} = GM\frac{1}{\vec{r}} \tag{12}$$

Since $\vec{U}$ depends on the mass density inside the Earth, it can be described as a function $U(\varphi, \lambda, r)$ of the latitude φ, the longitude λ, and the distance from the center of the Earth r. The mass M in the potential can be calculated taking the integral of the density over the entire volume of the Earth by using a series of Legendre polynomials to expand the integral. The result is a sum of elements depending on φ, λ, and r multiplied by coefficients depending on the mass distribution. The potential can be written as:

$$U = \frac{GM}{r}\sum_{n=0}^{\infty}\sum_{m=0}^{n}\frac{R^n}{r^n}P_{nm}(\sin\varphi)(C_{nm}\cos m\lambda + S_{nm}\sin m\lambda) \tag{13}$$

where P_{nm} is the Legendre polynomial of degree n and order m, and C_{nm} and S_{nm} are two coefficients that depend on the planet's internal mass distribution.

The gravitational potential is therefore an infinite sum of terms, each one describing a specific mass distribution model for the planet. For example, the first term of the series expansion is the potential associated to a perfectly spherical mass distribution, which is the same gravity acceleration considered in the two-body problem. The other terms of the series expansion represent *zonal* (varying with r and the latitude φ), *sectorial* (varying with r and the longitude λ), and *tesseral* (varying with r, λ, and φ) harmonics. The most relevant among these harmonics is the zonal harmonics of order 2, associated with Earth's *flattening* f, i.e., the Earth's radius at the equator being circa 20 km wider than the radius at the poles. This corresponds to $S_{20} = 0$ and a coefficient $C_{20} = -0.00108$ in the series expansion. This perturbation is also known as J_2, from the symbol used to identify the zonal coefficient $J_2 = -3/2C_{20} = 0.00162$. Other zonal harmonics of higher order are numerically less relevant.

The effects of the J_2 perturbation are mainly on Ω and ω. The extra mass at the equator produces an extra gravitational pull in the equatorial plane, which causes a precession of the orbit angular momentum. Depending on the altitude and the inclination of the orbit, the ascending node is shifted in the opposite direction of flight. The nodal regression rate caused by J_2 can be used to naturally achieve the necessary $\dot{\Omega}$ required for a Sun-synchronous orbit without making any maneuver, which is the case of many CubeSat missions. For example, at low altitudes, the orbit shall be almost polar to use the J_2 effect to achieve Sun synchronicity.

The effect of J_2 on ω is a shift in the orbit perigee, depending on the altitude and the inclination of the orbit. The apsidal rotation can be in the direction of flight or in the opposite direction, depending on the inclination.

3.2 Atmospheric drag

A spacecraft in low altitude orbit experiences an aerodynamic force due to the interaction between the atmosphere and the spacecraft surface. The main component of this force is the aerodynamic drag, while other components like lift force are usually negligible. The atmospheric drag is a force in the opposite direction to the spacecraft velocity v. The resulting acceleration $\ddot{\vec{r}}_{\text{drag}}$ is defined as:

$$\ddot{\vec{r}}_{\text{drag}} = -\frac{1}{2}\rho\frac{A}{m}v^2 C_D \hat{v} \tag{14}$$

where ρ is the atmospheric density (exponentially decreasing with the altitude), A is the spacecraft cross-sectional area, m is the spacecraft mass, and C_D is the drag coefficient. The primary effect of this perturbation is a decrease in the total energy of the satellite (nonconservative force). The force is opposed to the motion and makes the orbit's semimajor axis decrease. As a consequence, the orbital velocity increases (drag paradox). The increased speed further increases the value of drag, making the spacecraft spiral down. For elliptical orbits, the trajectory first becomes circular and then the orbital radius is reduced. The variation of the semimajor axis is

$$\dot{a} = -\frac{2a^2}{\mu}\ddot{\vec{r}}_{\text{drag}} m v \tag{15}$$

The atmospheric drag is very relevant for Earth orbits up to 1000 km, which are the most common for CubeSat missions. Above this altitude, the atmospheric drag can be neglected as other perturbations have more significant effects. Atmospheric drag can also be used in the approach to a generic celestial body with atmosphere to circularize the orbit. This concept, called aerobraking, is further described in Section 4.1.

The main consequence of atmospheric drag on a CubeSat mission is that of reducing the orbit lifetime. This is essential to respect the 25-year rule on spacecraft lifetime, which otherwise will be difficult to meet for CubeSats. Orbital decay due to atmospheric drag can be compensated with on-board propulsion systems. These aspects are dealt with in other chapters of this book and therefore will not be discussed further here.

Nevertheless, it is evident that the effects of atmospheric drag on the orbit need to be estimated during the mission design phase. This might be particularly difficult, because the variation of a depends on factors like the atmospheric density, the cross-sectional area, and the drag coefficient. The atmospheric density is estimated using atmospheric models, like the Harris-Priester density model [9] or the Jacchia density model [10], which of course are affected by errors and approximations.

The cross-sectional area depends on the attitude of the spacecraft at each instant. Since the time interval to be considered is quite large, A will vary and in general cannot be a priori predicted. Therefore an average value is used for the computation. The drag coefficient depends on the materials employed in spacecraft realization and the composition of the atmosphere and is therefore difficult to determine too. Even in this case, a reference value (usually around 2.2) can be adopted.

4 Leveraging natural dynamics in interplanetary missions

Among the critical challenges for micro- and nanosatellite interplanetary missions, propulsion is probably the most difficult since further technology advancement is physically limited by mass and energy constraints. It is therefore very important to maximally exploit a broad range of natural dynamical effects if a CubeSat-like spacecraft cannot be delivered directly to the nominal orbit around the target celestial body.

All the dynamical effects can be divided into two large groups depending on what type of fundamental interaction, gravitational or electromagnetic, underlies the effect. The first group effects can in turn be roughly split into two subgroups: those primarily based on two-body dynamics (the Oberth effect, GA maneuvers, resonant encounters) and essentially non-Keplerian phenomena commonly described by dynamical systems theory. The mechanisms and techniques of using solar radiation pressure and solar wind—such as solar sailing and electric sailing, respectively—comprise the second group. The gravitational dynamical effects are outlined next, with emphasis on their possible applications in CubeSat interplanetary missions.

4.1 GA maneuvers

The mechanism of GA maneuvers to reach a planet or a small body is conventional regardless of the spacecraft mass and size. For an unpowered (i.e., applying no thrust) GA maneuver, the intermediary's gravity field deflects the trajectory of a spacecraft entering its sphere of influence by an angle β related to the hyperbolic excess velocity V_∞ and the fly-by distance r_π (from the intermediary's center of mass) according to the formula:

$$\beta = 2\sin^{-1}\frac{\mu}{\mu + V_\infty^2 r_\pi} \tag{16}$$

The magnitude of the change in spacecraft heliocentric velocity after performing a GA maneuver can be calculated as follows:

$$\Delta V = 2V_\infty \sin\frac{\beta}{2} = \frac{2\mu V_\infty}{\mu + V_\infty^2 r_\pi} \tag{17}$$

The maximum velocity value attained is $V_{\max} = 2\sqrt{\mu/r_\pi}$ if $V_\infty^2 = \mu/r_\pi$. The largest achievable ΔV amounts to more than 30 km/s as a result of a GA maneuver near Jupiter for a spacecraft with radiation-hardened components. In a CubeSat mission, however, it can hardly exceed 15 km/s due to safety restrictions on the minimum fly-by distance. To obtain a higher Δv than the one generated by the GA, one could also consider an orbital maneuver performed at the periapsis of the orbit, where the energy gain is maximum. This is a consequence of the fact that the deeper a spacecraft is in the gravity well, the more efficient the propulsive force is. This effect, sometimes called the Oberth effect, is explained in further detail in Chapter 13 dedicated to Orbit Determination and Control.

To determine the feasible chains of intermediate GA maneuvers for multiple GA trajectories to a target celestial body, various analytical and semianalytical techniques have been developed. The most famous one is the Tisserand graph [11, 12]. It is based on the concept of Tisserand's parameter, a specific invariant combination of spacecraft orbital elements remaining constant after close encounter with an intermediary. Assuming the motion is planar, one can plot Tisserand's parameter contour lines on the plane periapsis distance versus orbital period. In some cases, it appears more convenient to use the plane periapsis distance versus apoapsis distance. The contour lines for different intermediaries (planets or moons) are then combined on a single plot—the Tisserand graph. The intersections indicate the potential feasibility of corresponding sequences. For essentially 3D spacecraft motion, the more complicated V_∞ globe-mapping technique is used [13].

Upon choosing a sequence of GA maneuvers, the trajectory optimization problem is then solved. During the past several decades, many automated trajectory design and/or optimization tools have been developed. Some of the tools have a built-in function of generating the optimal sequence. Various heuristic global optimization procedures are often used. An excellent review of the approved tools with their detailed descriptions can be found in Ref. [14].

The patching of two arcs before and after a GA maneuver can be improved by adding an impulse or a low-thrust arc during the intermediary fly-by. Such a GA maneuver is called "powered." The resulting hyperbola-to-hyperbola transfer problem was solved by Gobetz [15] who considered both optimal and periapsis single-impulse transfers. The latter, being easily calculated, are usually just slightly less efficient than the former. It is therefore this kind of powered GA maneuver that is available in most of the trajectory design tools. However, it is recommended not to include such an option in a CubeSat mission since it reduces the overall reliability by increasing the influence of GA maneuver execution errors and trajectory determination uncertainties.

It is also worth mentioning another extension of GA maneuvers: aerogravity assist (AGA) maneuvers [16–19]. For high lift-to-drag (L/D) ratio spacecraft, such as the so-called waveriders [20], the AGA maneuver gives an opportunity to bypass the otherwise unavoidable constraint on the turn angle resulting from the minimum fly-by distance limitations. This can be achieved by exploiting the negative lift force acting upon a waverider in the planet's atmosphere. Such a mechanism artificially increases the spacecraft weight and allows it to glide in the atmosphere at almost constant altitude for a much longer time as compared to a conventional GA maneuver. Bunches of efficient trajectories to the outer planets [18] and to the main belt asteroids [17] have been found. Similar to conventional GA maneuvers, AGA maneuvers near Venus can be effectively used for energy pumping, whereas Mars is best suited to perihelion changing [19].

One more promising application of the planet's atmosphere, the aerobraking effect, is available if the planet is the final destination. A spacecraft with low *L*/*D* ratio, being captured by the planet's gravitational field, can transfer from the initial highly elliptical orbit to the target near-circular low orbit by performing a series of fly-bys. The drag force described in the previous section can be exploited near the periapsis to gradually decrease the apoapsis altitude. This technique has been successfully tested in many missions near the Earth (Hiten, 1991), Mars (Mars Global Surveyor, 1997; Mars Odyssey, 2001; Mars Reconnasaince Orbiter, 2006), and Venus (Magellan, 1993; Venus Express, 2014). Unlike the aerocapture maneuver [21, 22] or skip reentry [23], aerobraking does not require superior thermal protection and is theoretically affordable for micro- and even nanosatellites equipped with deployable solar panels or a drag sail. To reduce the navigation-related risk, the use of autonomous closed-loop guidance, navigation, and control algorithms is recommended [24].

4.2 Resonant encounters

Consecutive GA maneuvers performed near the same intermediary are called resonant encounters. To ensure a series of resonant encounters near a planet or a moon, mid-course maneuvers (referred to as deep-space maneuvers, DSMs) are inserted into a heliocentric or, respectively, planetocentric arc of the trajectory. The efficiency of resonant encounters with a celestial body well suited for GA maneuvers was first revealed in the mid-1970s when the so-called ΔV-EGA maneuver—a series of GA maneuvers near the Earth—was discovered [25]. Later on, the term V_∞ leveraging maneuver (VILM) was coined for the corresponding DSM, and a solid theory has been developed [26, 27]. VILMs are performed when the spacecraft is at either the apoapsis (an exterior VILM) or the periapsis (an interior VILM) of its orbit. The magnitude of a VILM is relatively small but is enough for a significant increase/decrease in V_∞.

Resonant encounters and VILMs are particularly powerful tools in planetary moon tours, including the final phase of any tour, the endgame [28–30]. Basically, the endgame is a transfer to the target low orbit around the final destination moon. The complexity of the endgame phase consists of minimization of the fuel cost required for capture and lowering the orbit. In the patched conic approximation, the purely ballistic endgame with multiple fly-bys cannot lead to the decrease of the arrival speed V_∞ (the ballistic endgame paradox [29]). Therefore the use of VILMs is essential.

4.3 High-altitude fly-bys

The endgame design problem serves as an excellent example of modern trends in astrodynamics where the circular restricted three-body problem (CR3BP) model became dominant and, in many instances, replaced the patched conic approximation. The superiority of the CR3BP approach is explained by the fact that it reveals additional trajectory design opportunities undetectable when considering patched conics.

One of these opportunities is a high-altitude (distant) fly-by. In contrast to a conventional GA maneuver, the spacecraft "encounters" a planet/moon outside its sphere of influence. Such a low-energy encounter is impossible within the patched conic framework because it corresponds to the case $V_\infty^2 < 0$. Though a series of high-altitude fly-bys followed by lunar ballistic capture was utilized in the SMART-1 mission implemented as early as in 2003, a comprehensive theoretical basis for this technique was developed several years later [29, 31–36]. When applied to the endgame problem, high-altitude resonant encounters allow the ballistic endgame paradox to be solved. The elegant unified endgame theory extends the classic Tisserand graph to what is called the Tisserand-Poincaré graph [29], a convenient tool for efficient moon tour design [37].

Even more important are high-altitude resonant fly-bys for low-thrust transfers to the Moon similar to the SMART-1 spiral trajectory. Properly adjusted, distant lunar encounters are capable of saving much fuel by significantly increasing the geocentric orbit perigee [38, 39]. This is one of the most promising options for a piggyback CubeSat injected into either a low-Earth or the geostationary transfer orbit [40].

4.4 Weak stability boundaries and ballistic capture

The CR3BP model gives rise to a new type of transfer trajectory: low-energy trajectory. The action of a third body allows a spacecraft to arrive at the destination planet/moon with lower launch energy and fuel expenses. Probably the first example of a low-energy trajectory is the transit trajectory to the Moon suggested by Conley [41]. It terminates with lunar capture that does not require any insertion maneuver: the spacecraft-Moon Keplerian energy naturally changes sign from plus to minus due to the gravitational perturbation of a highly elliptical near-Earth orbit. This is what Belbruno later called ballistic capture [42, 43]. In contrast to capture by means of propulsion, ballistic capture has proved to be temporary. It is therefore also referred to as weak.

To study low-energy trajectories in the three-body or even four-body system (e.g., Earth-Moon-Sun-spacecraft), Belbruno introduced an extension of the sphere of influence concept, termed a weak stability boundary (WSB) [42, 43]. The original algorithmic definition of the WSB as a boundary of the region in the configuration space where the spacecraft motion about one of the massive celestial bodies is stable (i.e., the spacecraft makes at least one revolution about this body) was subsequently supplemented with a more general and rigorous analytic definition [32, 33]: the extended WSB of a body is set in the phase space lying in the intersection of a given Jacobi integral hypersurface and the hypersurface of zero Keplerian energy (relative to the body in hand). In the planar CR3BP case, Poincaré map and Keplerian (periapsis) map analysis reveal the chaotic nature of ballistic capture and its close connection with the WSB and transitions between the resonant orbits. The resonance hopping theory [29, 31–36], mentioned earlier as applied to high-altitude resonant encounters, treats resonant transitions as hops in the chaotic "sea" surrounding the stable resonance "islands." The intrinsic link between the WSB and the stable invariant manifolds

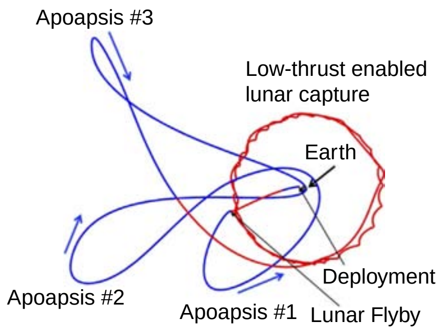

Fig. 3 The baseline Lunar IceCube trajectory [46]. Coast arcs are indicated by *blue* (*dark gray* in print version); low-thrust arcs are *red* (*light gray* in print version). The figure is plotted in the Sun-Earth rotating frame. The Sun is from the left.

associated with libration point orbits has also been discovered [44, 45]. A fruitful field of space manifold dynamics will be addressed in the following subsection.

Among the practical achievements of the WSB theory, one can point out two successful missions to the Moon, Hiten (1991) and GRAIL (2011), designed using the algorithmic WSB definition. When launched toward the Sun-Earth L1 libration point, a spacecraft experiences the Sun's perturbation force, which deflects the trajectory to the Moon's vicinity, where ballistic capture occurs. A similar idea is used in the Lunar IceCube project. Scheduled for 2020, this groundbreaking CubeSat mission to the Moon includes lunar capture preceded by a low-thrust arc and a long WSB-type transfer phase (Fig. 3).

Low-energy trajectories to the Moon and the Solar System planets, especially those ending with ballistic capture, have many useful benefits and have the only drawback of the long duration of the cruise phase. The benefits include

1. The reduction in ΔV required (typically from 10% to 30%).
2. The elimination of the orbit insertion maneuver, usually the riskiest and most expensive in terms of fuel maneuver in the whole mission.
3. Significantly larger launch windows. At the same time, the arrival date can often be fixed for any start date from the launch window.

All these advantages become increasingly important for low-cost and medium-cost small spacecraft with limited launch opportunities and propulsion capabilities (particularly for piggyback CubeSats with low-thrust propulsion systems).

4.5 Hyperbolic manifolds and interplanetary transport network

The crucial topological entities in the CR3BP model are periodic and quasiperiodic orbits near the collinear libration points (CLPs) and the hyperbolic invariant manifolds, stable and unstable, associated with these orbits. Among all five relative equilibria in the rotating (synodic) reference frame, the CLPs, denoted by L_1, L_2, and L_3 (Fig. 4), are unstable for any mass ratios m_2/m_1 of the primaries. The phase space in the

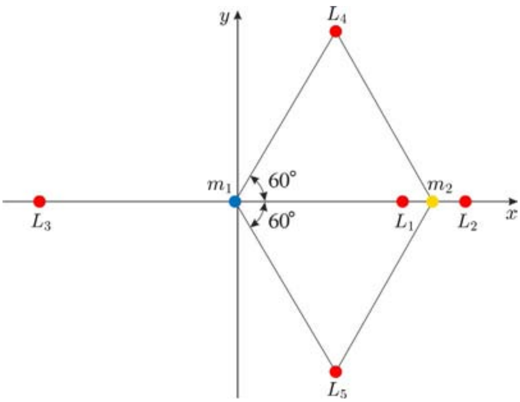

Fig. 4 Libration points as relative equilibria in the CR3BP dynamical model.

close vicinity of CLPs is of saddle × center × center type. The 4D center × center manifold is formed by (quasi)periodic orbits. The saddle component manifests as two asymptotic trajectories emanating from each point of any unstable (quasi)periodic orbit, one from the stable manifold and another from the unstable manifold (Fig. 5). Hyperbolic invariant manifolds of different libration point orbits in the same

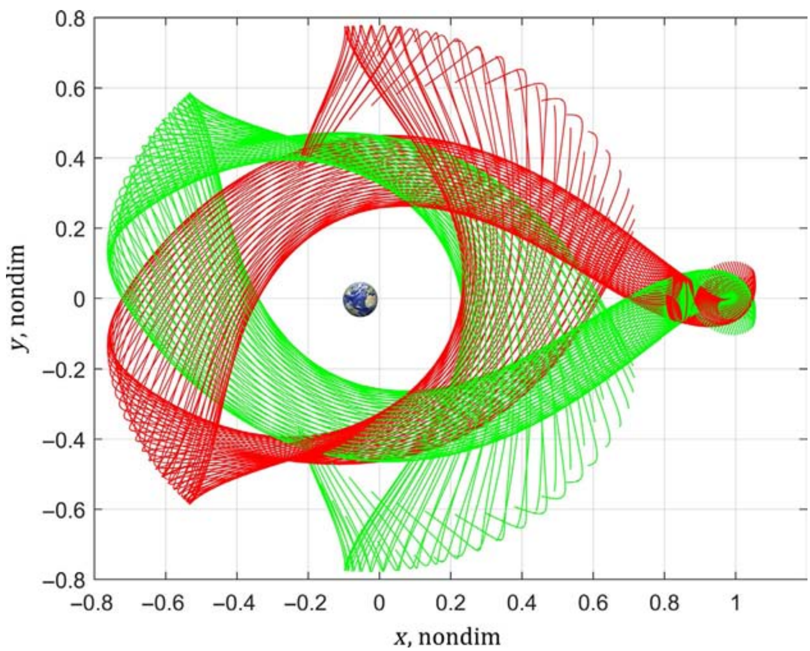

Fig. 5 Stable [*green* (*light gray* in print version)] and unstable [*red* (*dark gray* in print version)] invariant manifolds emanating from some L_1 halo orbit in the Earth-Moon system. The Earth is not drawn to scale. The Moon is hidden by hyperbolic invariant manifolds.

three-body system or even different three-body systems can intersect and, similar to ocean currents, serve as a propellant-less deep-space transport mechanism.

A turning point in astrodynamics was marked at the end of the 1990s when, for the first time, dynamical systems theory (in particular, invariant manifold dynamics) was applied to the mission design [47, 48]. Based on early fundamental investigations [41, 49], the space manifold was soon developed and successfully exploited in designing the Genesis mission [50, 51]. The summit of the manifold approach success was probably the proposed idea of the so-called Petit Grand Tour between the Jovian moons [52–54] and the most remarkable discovery—the existence of the vast Solar System network consisting of stable and unstable invariant manifolds of the planets and their moons, called the Interplanetary Superhighway (IPS) [55], or Interplanetary Transport Network (ITN) [56]. Chaotic regions close to collinear libration points play the role of "portals" to the "tunnels" confined by hyperbolic manifold "walls." For the outer planets, the stable and unstable manifolds of any two neighboring planets intersect, allowing a spacecraft to migrate across the Solar System with little or no fuel.

To the present day, astrodynamicists all over the world continue generating a substantial amount of literature related to the design of high-thrust and low-thrust interplanetary transfers augmented with hyperbolic manifold arcs. In the context of CubeSat missions, such trajectories are especially promising provided the short lifetime problem is solved for electronics and propulsion system components.

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